the only substantive function of a space station, as far as I can see, is for long-duration spaceflight. How do humans behave in microgravity? How can we counter progressive changes in blood chemistry and an estimated 6 percent bone loss per year in zero gravity? (For a three- or four-year mission to Mars this adds up, if the travelers have to go at zero g.)

These are hardly questions in fundamental biology such as DNA or the evolutionary process; instead they address issues of applied human biology. It's important to know the answers, but only if we intend to go somewhere in space that's far away and takes a long time to get there. The only tangible and coherent goal of a space station is eventual human missions to near-Earth asteroids, Mars, and beyond. Historically NASA has been cautious about stating this fact clearly, probably for fear that members of Congress will throw up their hands in disgust, denounce the space station as the thin edge of an extremely expensive wedge, and declare the country unready to commit to launching people to Mars. In effect, then, NASA has kept quiet about what the space station is really for. And yet if we had such a space station, nothing would require us to go straight to Mars. We could use a space station to accumulate and refine the relevant knowledge, and take as long as we like to do so —so that when the time does come, when we are ready to go to the planets, we will have the background and experience to do so safely.

The *Mars Observer* failure, and the catastrophic loss of the space shuttle *Challenger* in 1986, remind us that there will be a certain irreducible chance of disaster in future human flights to Mars and elsewhere. The *Apollo 13* mission, which was unable to land on the Moon and barely returned safely to Earth, underscores how lucky we've been. We cannot make perfectly safe autos or trains even though we've been at it for more than a century. Hundreds of thousands of years after we first domesticated fire, every city in the world has a service of firefighters biding their time until there's a blaze that needs putting out. In Columbus' four voyages to the New World, he lost ships left and right, including one third of the little fleet that set out in 1492.